

Practice Contents

1. A *relative path* starts at the current directory. An *absolute path* starts at the Linux root level, / as in */home/foxr/myfiles*. NOTE: remember to use tab completion when possible!

a. Enter **cd /**.

- *What does this command do?*

Chuyển về thư mục gốc (root directory)



```
HE210528@OSG203:~$ pwd
/home/HE210528
HE210528@OSG203:~$ cd /
HE210528@OSG203:/$ $
```

Type **cd etc** followed by **cd sysconfig** and **cd networking**. Type **pwd**.

- *Where are you?*

/etc/sysconfig/networking

Type **cd /etc/sysconfig/network-scripts**. Type **cd ../../**

- *Where are you?*

thư mục /etc

Type **cd ~**.

- *Where are you?*

Đưa về thư mục Home directory của tài liệu người dùng hiện tại đang đăng nhập

Type **cd ~Student**.

- *Where are you?*

Đưa đến thư mục /home/Student

- b. From Student's home directory, type **ls**, one item is the subdirectory **FILES** which has a subdirectory called **DUMMY-DIRECTORY**. Enter a command to **cd** into **DUMMY-DIRECTORY** using a *relative* path.

- *What command did you enter?*

cd FILES/DUMMY-DIRECTORY

- *What directory are you in according to your prompt?*

Thư mục hiện tại trên dấu nhắc (prompt) sẽ hiển thị là DUMMY-DIRECTORY

- c. We want to return to the network-scripts subdirectory (see 1a) from **DUMMY-DIRECTORY**. Enter a **cd** command to do so using an *absolute* path.

- *What command did you enter?*

cd /etc/sysconfig/network-scripts

- *From here, what command would you enter using an absolute path to return to Student's DUMMY-DIRECTORY?*

`cd /home/Student/FILES/DUMMY-DIRECTORY`

- d. Your PATH variable was defined in `/etc/profile` and appended to in `/etc/bashrc` and in again in `.bashrc_profile`.

- What are the contents of PATH?

`echo $PATH`

Hint: recall how to output the value of an variable from lab 1 step 6.

To discover where a command is, we can use `which` and `whereis`, as in **which wc** or **whereis wc**. *which* will only search for the executable program in directories stored in your PATH variable and *whereis* will search not just for the executable but the man page and other related files.

- Where is `wc` stored?

`/usr/bin/wc`

- Where is `ip` stored?

`/usr/sbin/ip`

- e. Type **which systemd**.

- What response did you get?

`/usr/lib/systemd` không nằm trong biến môi trường PATH

Type `PATH=$PATH:/usr/lib/systemd`. Now retry **which systemd**.

- What response do you get? Explain the assignment statement (`PATH=$...`).

Lệnh `which systemd` sẽ tìm thấy file tại đường dẫn mới

2. Exploring `touch`, `cp`, `mv`, `rm`, `rmdir`. Remember tab completion in steps like 2b, 2c.

- a. `cd` to your home directory. Type `touch test1.txt test2.txt test3.txt test4.txt` to create these four files. Type `ls -l`.

- What are the sizes for these four files?

Size của cả 4 tệp tin này đều bằng 0

- b. `cd` to `~Student`. Let's copy all `.txt` file in **FILES** to your home directory.

- What command did you enter?

`cp ~Student/FILES/*.txt ~`

Type `cd ~`. To indicate that files should be copied here (your current directory), the destination directory is denoted using `.` (period). Enter a `cp` command to copy the files in Student's DUMMY-DIRECTORY/directory2 to your current directory.

- What command did you come up with?

`cp ~Student/DUMMY-DIRECTORY/directory2/* .`

- c. From your home directory, type `echo hi >> aa1.abC`. Type `ls -l` and you will see that `aa10.bBC` is still empty while `aa1.abC` has a size of 3. Repeat the last `cp` command from step 2b which copied those two files to this directory but add the option `-i` (interactive mode). Answer `n` when asked. Repeat the `ls -l` command and you will see nothing changed. Repeat the `cp -i` command but this to answer `y`. Repeat `ls -l` and you will see that `aa1.abC` is back to a size of 0. Repeat the `cp` command without the `-i` option.

- What happens? Explain why and when you should use `-i` with `cp`.

Khi không có tùy chọn `-i`, lệnh `cp` mặc định sẽ ghi đè lên các file trùng tên mà không thông báo. Tùy chọn `-i` buộc hệ thống hỏi lại người dùng để xác nhận (y/n) trước khi ghi đè dữ liệu

- d. Type the command `cp -r ~Student/DUMMY-DIRECTORY/directory3 temp`. You will receive a few permission denied errors of items that you do not have access to copy, ignore those. Type `ls ~Student/DUMMY-DIRECTORY/directory3` and then `ls temp` and compare what you find.

- *In your own words, what does recursive mean?*

"Recursive" có nghĩa là lệnh sẽ được áp dụng cho toàn bộ thư mục chỉ định, bao gồm thư mục đó, tất cả các thư mục con bên trong nó, và tất cả các tệp tin nằm trong các thư mục con đó xuyên suốt cấu trúc cây thư mục

- e. With the **temp** directory created from 2d, **cd** into **temp** and type **mv DUMMY-DIRECTORY temp2**.
- *What does this instruction do? Is **temp2** there? What about **DUMMY-DIRECTORY**?*

Type **mv temp2 ..**. Type **ls** and then **cd ~** and **ls**.

- *What did this second **mv** command do?*

Lệnh **mv temp2 ..** thực hiện di chuyển thư mục **temp2** từ vị trí hiện tại lùi lại một cấp

- f. In one **cd** command, move to **temp2** (up 1 level and then down into **temp2**).

- *What command did you issue?*

cd ../temp2

Type **ls** and you should find 4 files ending in **.dat**. Enter **mv *.dat *.txt**.

- *What happened?*

Hệ thống hiển thị thông báo lỗi và các file **.dat** không hề được đổi tên sang **.txt**

Type **mv file1.dat file1.txt**.

- *This version works by renaming one file so why didn't **mv** work using the ***** to rename all 3 files?*

Lệnh **mv** không thể hoạt động với ký tự đại diện ***** để đổi đuôi hàng loạt

Using command line editing, repeat this last **mv** command 3 times to rename the other 3 files to have **.txt** extensions.

- g. In step 2b, you copied **sales.txt** to your home directory. Type **cd ..** followed by **rm sales.txt**. Type **ls** to see if the file was deleted. Move back into **temp2** where there are several **.dat** files. Enter a command to delete these **.dat** file using a single **rm** instruction and appropriate wildcards (only delete the **.dat** files).

- *What command did you enter?*

rm *.dat

- h. While still in **temp2**, type **rm -i *.txt**. Answer **n** when prompted.

- *How could you ensure that whenever you type **rm**, the **-i** option is used without having to necessarily type **rm -i**?*

Tạo một **alias** (biệt danh) bằng cách thêm dòng sau vào tệp cấu hình **~/.bashrc**
alias rm='rm -i'

- i. One subdirectory of **DUMMY-DIRECTORY** (which is now **temp2** in your directory) is **dir5**. This is an empty directory. Type **rm dir5**.

- *What happened?*

Retry the delete command using **rmdir**.

- *What command did you use? Did it work?*

Now try to delete the directory **directory2**.

- *What response did you get?*

cd into **directory2** and type **rm ***. Now **cd** up 1 level and repeat the **rmdir** command.

- *Did it work?*

cd up 1 level to **~** and type **rm -r temp2**.

- *Type **ls** and explain what the **rm** command did.*

- j. The opposite of **rmdir** is **mkdir**, to create a new directory. From your home directory, type **mkdir new**. Type **ls -l**.

- *What size is this directory (even though it is empty)?*
Kích thước hiển thị thường là 4096 bytes
- *Do its permissions (the first 10 characters of the line) match the other directories? If not, how does it differ?*
Có, trùng khớp. Quyền hạn của nó bắt đầu bằng ký tự d

k. The files *aa10.bBC*, *aa1.abC*, and 3 *.txt* files were all copied from *Student*. Use a single *mv* command with wildcards to move them all into *new*.

- *What command did you use?*
`mv aa*.bBC aa*.abC *.txt new/`

3. The commands *head/tail* display the first and last parts of a file, let's explore them.

a. *cd* to */etc*. Type **cat passwd**, **head passwd** and **tail passwd**.

- *How did the three commands' outputs differ?*
 - + **cat**: **Hiển thị toàn bộ nội dung file**
 - + **head**: **Chỉ hiển thị 10 dòng đầu tiên**
 - + **tail**: **Chỉ hiển thị 10 dòng cuối cùng**

b. Type **tail -c 100 passwd**.

- *What does this command do?*
`tail -c 100 passwd`: **Hiển thị 100 byte cuối cùng của file**

Now try **head -n 5 passwd**.

- *What does this command do?*
`head -n 5 passwd`: **Hiển thị 5 dòng đầu tiên của file**

c. Try these out: **head -n -10 passwd** and **tail -c +100 passwd**.

- *Explain what `-n -10` and `-c +100` do in these cases.*
 - + **`-n -10` (trong head)**: **Hiển thị toàn bộ file, bỏ qua 10 dòng cuối cùng**
 - + **`-c +100` (trong tail)**: **Hiển thị nội dung file bắt đầu từ byte thứ 100 trở đi đến hết file**

d. *What command would list the passwd file starting at line 6?*

`tail -n +6 passwd`

e. While you are in */etc*, type **file a***.

- *What types of entities do you find?*
Thư mục directory, file văn bản ASCII text, và liên kết mềm symbolic link

Repeat this step using the option `-i`.

- *How does the information about the file type differ from the version without using `-i`? (examine file's man page if necessary)*
 - + **`Không dùng -i`**: **Hiển thị mô tả định dạng dễ đọc bằng ngôn ngữ tự nhiên (Ví dụ: ASCII text, directory)**
 - + **`Dùng -i`**: **Hiển thị định dạng chuẩn MIME-type của hệ thống (Ví dụ: text/plain; charset=us-ascii, application/x-directory)**

4. More complex file commands: *cmp*, *diff*, *join*, *paste*, *cut*, *split*.

a. *cd* to your directory *new* in your home directory. Type **cat << quit > temp1.txt** and enter a list of 5 first names. Repeat this to create *temp2.txt* with the same list of names except that *temp2.txt* should omit the fourth name entirely and misspell the second name.

Type **cmp temp1.txt temp2.txt**.

- *What is the output?*
`temp1.txt temp2.txt differ: byte 7, line 2`

Repeat using diff.

- *What is diff's output?*
Chỉ ra cụ thể các dòng khác nhau (dòng 2 bị thay đổi, dòng 4 và 5 bị xóa ở file 2)
- Read diff's man page and then *explain precisely what the output from the diff command means.*
< là dòng chỉ có ở file 1; > là dòng chỉ có ở file 2. Các chữ c (sửa), d (xóa) là hành động cần làm để đồng bộ file 1 giống file 2

- b. Now issue two more *cat* commands similar to those of 4a except that the first list of names should precede each name with a number from 1-5 as in `1. Frank` and for the second file, do the same but last names, again, numbered 1-5, like `1. Zappa`.

Now issue a *join* command where the command receives both file names, as in **join temp3.txt temp4.txt**.

- *What is the result? Explain what the join command does.*
1 Frank Zappa

A related command is *paste*; issue the paste command on the same two files.

- *How does the output differ?*
paste ghép song song 2 file theo chiều ngang mà không so khớp, giữ nguyên cả hai cột số thứ tự (1 Frank 1 Zappa)

- c. The *cut* command allows you to cut and return part of the content of a file. One way to control cut is to supply the field(s) you want returned. In the case of the two files you created in 4b, we want to return the names but not the numbers. This would be field 2. The command would be **cut -f 2 filename**. However, cut expects fields to be separated by tabs instead of spaces. We need to modify the command to use a space as a delimiter instead of a tab, using **-d ' '** (1 blank space). This should then return all of filename without the numbers or space before each name. Try it. The output is sent to the window. Issue two separate cut commands, one for each of the files from 4b, redirecting the output to two new files. Now issue the paste command on those two files. Repeat the same command but using join.

- *How does the output of paste differ from the output of paste from the end of 4b?*
Chỉ còn cột tên và họ ghép sát nhau (Frank Zappa), không còn cột số thứ tự ở giữa
- *How does the output of join on these two files differ from the join command from 4b when you used the original files?*
Kết quả trống (không hiển thị gì) vì sau khi cut, hai file không còn trường chung nào để đối khớp.

- d. One file you had copied earlier is called addresses.txt which is in this current directory. The syntax is **split -l # file prefix** which splits file into multiple files of # lines naming each file using prefix as the name of each file but adding to each file a two-letter or two-digit value like aa or 01.

We will split the original file into files of 3 addresses each, using **-l 3**, and with the prefix **addr**.

- *What command did you enter?*
split -l 3 addresses.txt addr
- *What new files were created?*
addraa, addrab, addrac...

Repeat your command but add the option **-d** after **-l 3**.

- *How are these new files named?*
addr00, addr01, addr02...
- *What does -d do?*
Sử dụng hậu tố bằng chữ số thay vì chữ cái.

Delete all of these newly created files, including addresses.txt (you can delete them all with one rm command using wildcards).

5. Links provide easy access to other files or moved content. The two types of links are hard links and symbolic links.

- a. Type **ls -l ~Student/FILES**. The second item in the listing is the number of file records (hard links) pointing at the given file.

From ~Student, enter the command **ln -s FILES/addresses.txt addresses**. Type **ls -l** and you should see a new entry for addresses.

- *What is this new item?*

Là một **symbolic link** (liên kết mềm) trỏ tới file gốc

It is not a copy of the file but instead a symbolic link. Again, type **ln -l ~Student/FILES** and compare what you see for *addresses.txt* with what you see for *addresses*.

- *Specifically, how do the first and last columns of the long listing of both items differ?*

+ **Cột đầu tiên (Quyền hạn):** File liên kết bắt đầu bằng chữ **l** (**lrwxrwxrwx**), còn file gốc bắt đầu bằng dấu **-** (**-rw-r--r--**).

+ **Cột cuối cùng (Tên):** File liên kết hiển thị mũi tên chỉ đường dẫn *addresses -> FILES/addresses.txt*, còn file gốc chỉ hiển thị tên file *addresses.txt*

Type **cat addresses**.

- *Does it display the contents of the file?*

Có

- b. Type **ln sales.txt sales**. This creates a hard link and inserts it in the same directory.

Type **ls -l** and *compare how these two entries differ (if at all)*.

Không khác nhau về quyền hạn và kích thước, nhưng số lượng liên kết cứng ở cột thứ 2 của cả hai file đều tăng lên thành 2

- c. Delete *sales.txt* and type **ls -l**.

- *What has changed?*

File *sales.txt* bị xóa, và số liên kết ở cột thứ 2 của file *sales* giảm từ 2 xuống 1

Type **cat sales**.

- *Did it work?*

Có (vì liên kết cứng lưu inode trực tiếp của dữ liệu nên file vẫn đọc được)

Type **su Student** to switch to *Student*, the password is *LinuxIsCool!*, and **cd** to *~Students/FILES*.

Type **cp addresses.txt addresses2.txt** and then delete *addresses.txt*.

Type **exit** to return to your own account. Type **ls -l**

- *Explain what has happened to the addresses symbolic link.*

Liên kết bị hỏng (**broken link**) và nhấp nháy đỏ trên terminal vì file gốc *addresses.txt* không còn tồn tại

Delete the symbolic link.

- *How did you do that?*

Sử dụng lệnh: **rm addresses**

su back to *Student*, return to the *FILES* directory and **mv addresses2.txt** to *addresses.txt* to retain a copy.

Close your terminal window and shut down your VM.